

VirtualWorks Pharmacovigilance Internship

Task 3 – Causality Assessment

Prepared by: Nagesh Dongare

Method: WHO-UMC causality assessment

Document type: Educational simulated case analysis

1. Objective

To evaluate the relationship between a suspected medicine and an adverse reaction using the WHO-UMC causality assessment approach and classify cases as **Certain, Probable/Likely, or Possible**.

2. WHO-UMC Criteria Used

Category	Key assessment points
Certain	Plausible time relationship; cannot be explained by disease or other drugs; clinically plausible response to withdrawal; even
Probable / Likely	Reasonable time relationship; unlikely to be explained by disease or other drugs; clinically reasonable improvement after v
Possible	Reasonable time relationship, but the event can also be explained by disease or another drug; dechallenge information m

3. Case Study 1 – Classified as CERTAIN

Case ID: SIM-CAUS-001

Patient: 28-year-old female

Suspected drug: Ceftriaxone injection

Indication: Bacterial infection

Reaction: Anaphylaxis (hypotension, wheezing and facial swelling)

Timeline: Symptoms began within minutes of administration. The patient had received ceftriaxone previously.

Action/outcome: Ceftriaxone was stopped and emergency treatment was given; symptoms improved.

Assessment: The onset is highly plausible and immediate. The clinical picture is a specific serious allergic phenomenon with a strong temporal relationship. No alternative cause was identified. Rechallenge was not attempted because deliberate re-exposure after suspected anaphylaxis would be inappropriate.

Final classification: **CERTAIN**.

4. Case Study 2 – Classified as PROBABLE / LIKELY

Case ID: SIM-CAUS-002

Patient: 45-year-old male

Suspected drug: Amoxicillin 500 mg orally three times daily

Indication: Bacterial throat infection

Reaction: Generalized diarrhea

Timeline: Diarrhea started about 2 days after amoxicillin was initiated.

Action/outcome: Amoxicillin was discontinued after clinical review; diarrhea improved over the next few days. No other new medicines were identified and no evidence of a gastrointestinal infection was reported.

Assessment: There is a reasonable temporal relationship. The reaction is a known type of adverse effect associated with antibiotics. Alternative explanations were not strongly supported, and improvement after withdrawal was clinically reasonable. Rechallenge was not required.

Final classification: **PROBABLE / LIKELY**.

5. Case Study 3 – Classified as POSSIBLE

Case ID: SIM-CAUS-003

Patient: 60-year-old female

Suspected drug: Metformin 500 mg twice daily

Indication: Type 2 diabetes mellitus

Reaction: Dizziness

Timeline: Dizziness was reported after metformin was started. The patient was also taking an antihypertensive medicine and had variable blood-pressure readings.

Action/outcome: No clear dechallenge information was available.

Assessment: The timing provides a reasonable possibility of a relationship with metformin, but dizziness could also be explained by the concomitant antihypertensive treatment or blood-pressure changes. Because alternative explanations are plausible and dechallenge is unclear, the evidence is insufficient for Probable/Likely.

Final classification: POSSIBLE.

6. Summary Table

Case	Drug	Reaction	Classification	Main reason
SIM-CAUS-001	Ceftriaxone	Anaphylaxis	Certain	Immediate, specific clinical phenomenon; strong timing; no alternative causes
SIM-CAUS-002	Amoxicillin	Diarrhea	Probable / Likely	Reasonable timing, alternative causes unlikely, improvement after withdrawal
SIM-CAUS-003	Metformin	Dizziness	Possible	Reasonable timing, but other plausible causes exist and dechallenge is unclear

7. Conclusion

Causality assessment is a structured clinical judgement rather than proof of causation. Timing, alternative causes, dechallenge, clinical characteristics and the quality of available information should be considered. These simulated cases demonstrate how the WHO-UMC approach can lead to Certain, Probable/Likely and Possible classifications.

8. Originality and Educational Note

These are original simulated educational cases prepared for internship practice. They are not real patient records and do not reproduce a published patient case. They are intended to demonstrate the assessment method and should not be used as clinical advice.

9. References

- World Health Organization. The use of the WHO-UMC system for standardised case causality assessment.
- Uppsala Monitoring Centre. Summary description of Causality Assessment.
- Pharmacovigilance Programme of India (PvPI). Guidance Document for Spontaneous Adverse Drug Reaction Reporting.